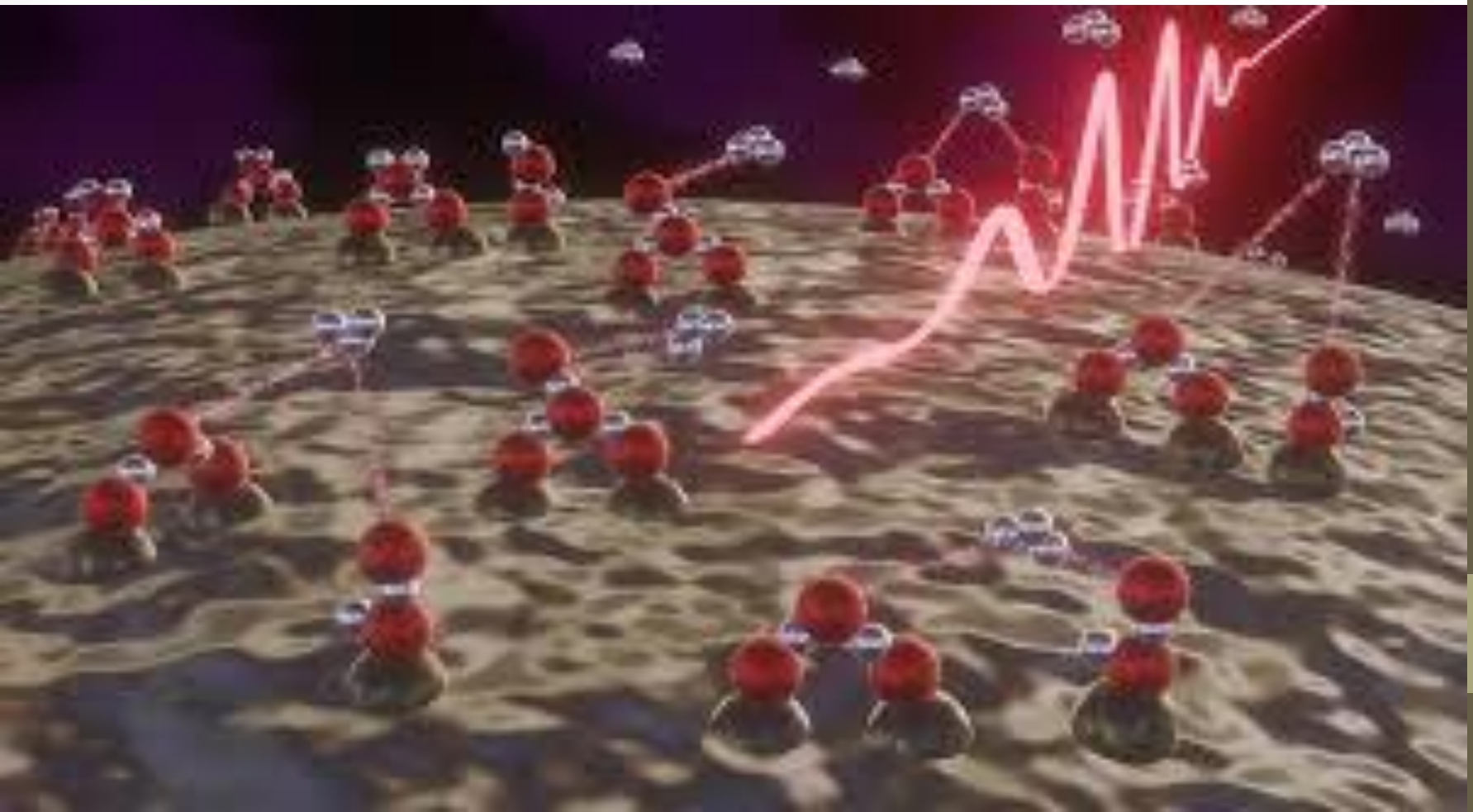


NATURE OF MATTER



Black Body Radiation:

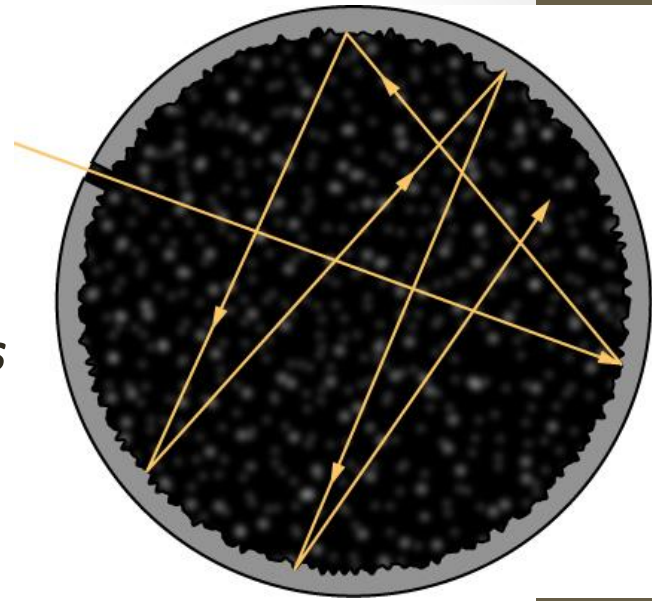
A blackbody is defined as,

“An ideal body that allows all incident radiation to pass into it (zero reflectance) and that absorbs internally all the incident radiation (zero transmittance) ,regardless of its frequency, is called a Black Body.”

A blackbody emits blackbody radiation which is also known as thermal radiation.

A good example of a black body is a cavity with a small hole in it. Any light incident upon the hole goes into the cavity and is essentially never reflected out since it would have to undergo a very large number of reflections off walls of the cavity.

The walls are absorptive (perhaps by painting them black), the cavity makes a perfect black body.



Characteristics:

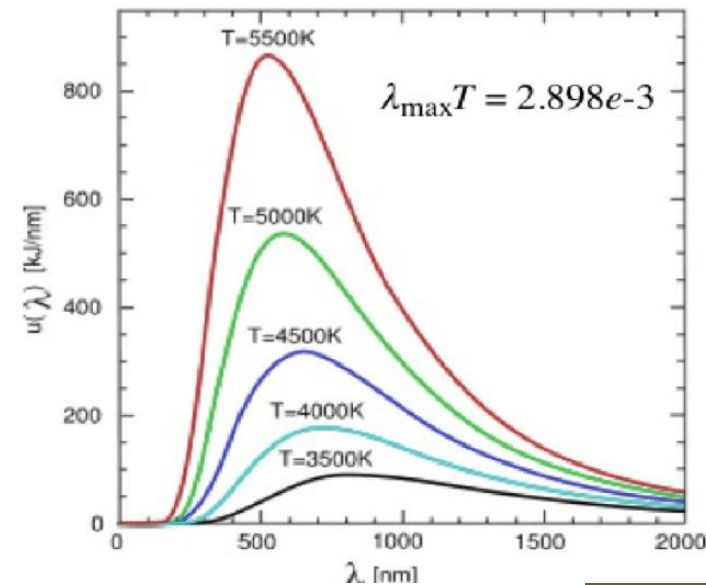
1. Such radiation is independent of the material of a black body.
2. Such radiation is only depend on the temperature of the cavity.
3. Such radiation is lie in the region of infrared and visible light only.
4. Such radiation is a group of different wavelength at a particular temperature.

Wien's Displacement Law

Wien's displacement law states that,

The blackbody radiation curve for different temperatures peaks at a wavelength inversely proportional to the temperature.

$$\begin{aligned}\lambda_{\max} &= \frac{\text{constant}}{T} \\ &= \frac{2.90 \times 10^{-3}}{T}\end{aligned}$$



The shift of that peak is a direct consequence of the Planck radiation law which describes the spectral brightness of black body radiation as a function of wavelength at any given temperature.

The entire shift of the spectrum of black body radiation toward shorter wavelengths as temperature increases.

Stefan-Boltzmann Law

The thermal energy radiated by a blackbody radiator per second per unit area is proportional to the fourth power of the absolute temperature

$$\frac{P}{A} = \sigma T^4 \text{ } j / m^2 s \quad \text{Stefan-Boltzmann Law}$$

$$\sigma = 5.6703 \times 10^{-8} \text{ watt } / m^2 K^4$$

For hot objects other than ideal radiators: $\frac{P}{A} = e\sigma T^4$

where, e is the emissivity of the object (e = 1 for ideal radiator). If the hot object is radiating energy to its cooler surroundings at temperature T_c , the net radiation loss rate takes the form

The Stefan-Boltzmann relationship is also related to the energy density in the radiation in a given volume of space.

Rayleigh-Jeans Law

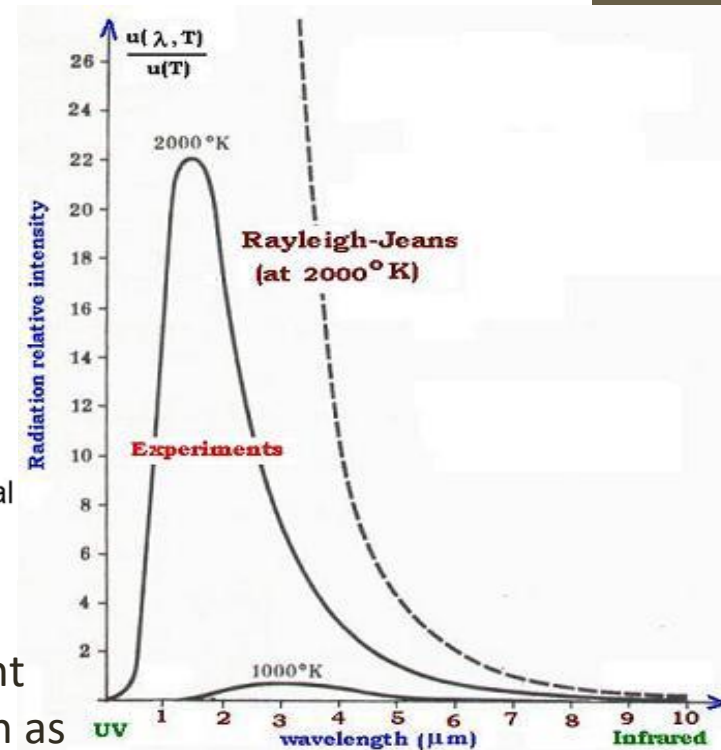
It is an approximation in respect to thermal radiation:

The emissive power of a blackbody at absolute temperature T and at a given wavelength λ is directly proportional to T and inversely proportional to λ^4 .

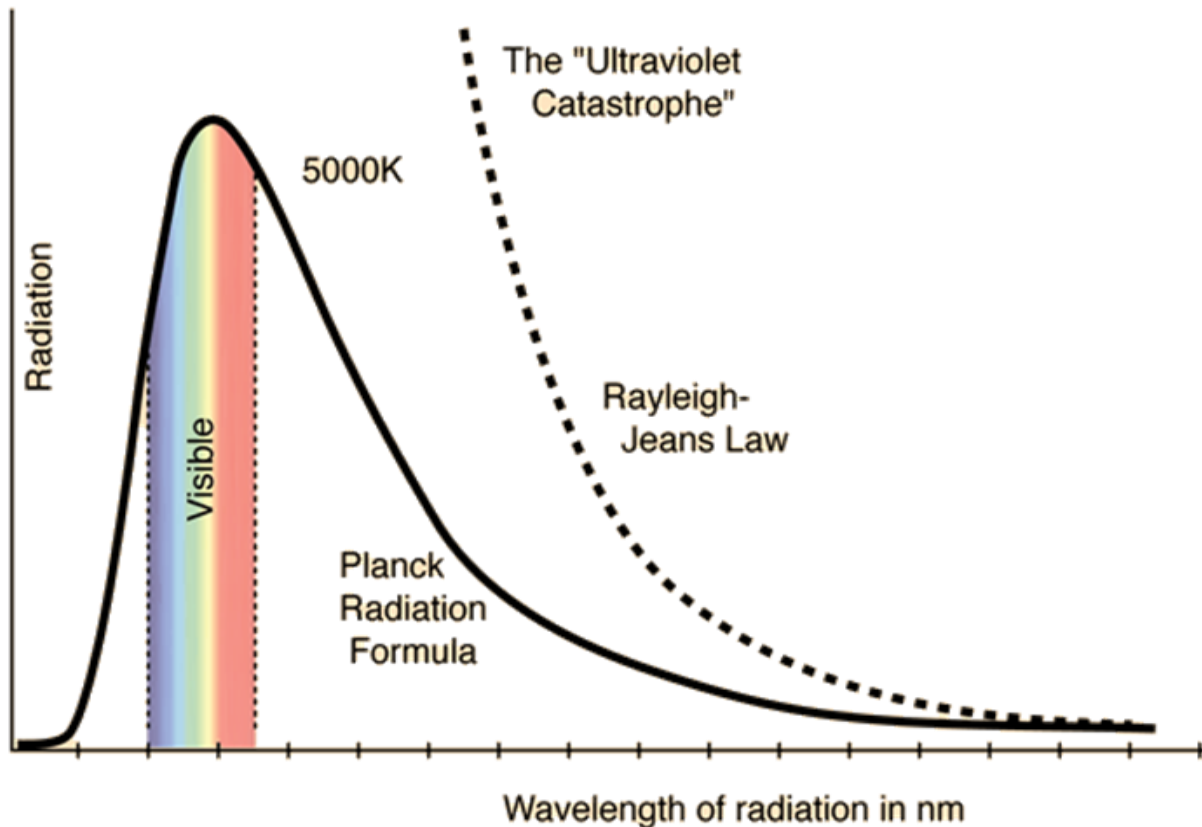
$$B_{\lambda}(T) = \frac{2ck_B T}{\lambda^4}$$

- Classical prediction of radiance from a blackbody at temperature T
- c : speed of light ($2.998 \times 10^8 \text{ m s}^{-1}$)
- k_B : Boltzmann's constant ($1.381 \times 10^{-23} \text{ J K}^{-1}$)
- Blackbody emission proportional to temperature, and inversely proportional to the fourth power of wavelength

At shorter wavelengths there was a major disagreement with the experimental data and this mismatch is known as ultraviolet catastrophe.



As it occurred in the ultraviolet portion of the spectrum and stated that at low wavelengths the intensity should become infinite, this obviously was a violation of the law of conservation.



Plank's Quantum Explanation of Black Body Radiation:

According to Plank's quantum theory,

- Different atoms and molecules can emit or absorb light in discrete quantities only.
- The smallest amount of energy that can be absorbed or emitted is an E.M.R known as quantum.
- The energy of radiation **absorbed** or **emitted** is directly proportional to the frequency of radiation.

$$E = hf$$

Where,

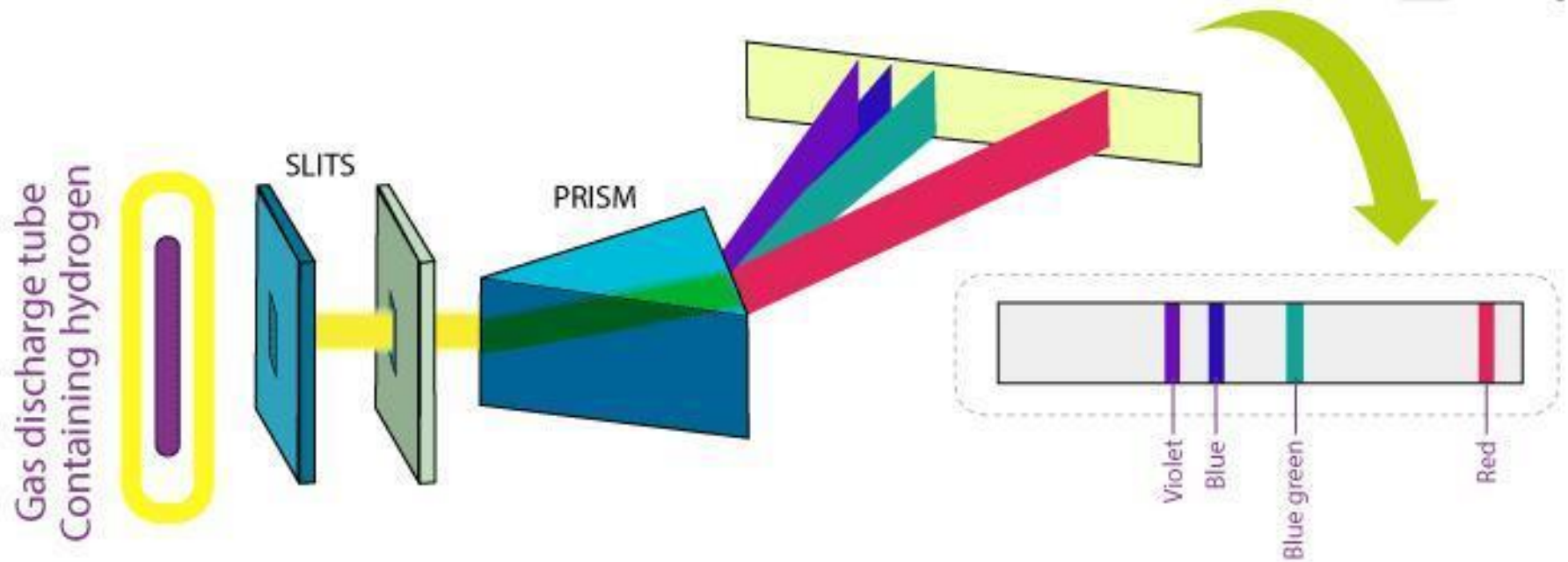
E = energy of radiation

h = plank's constant (6.63×10^{-34} J.s)

f = frequency of radiation

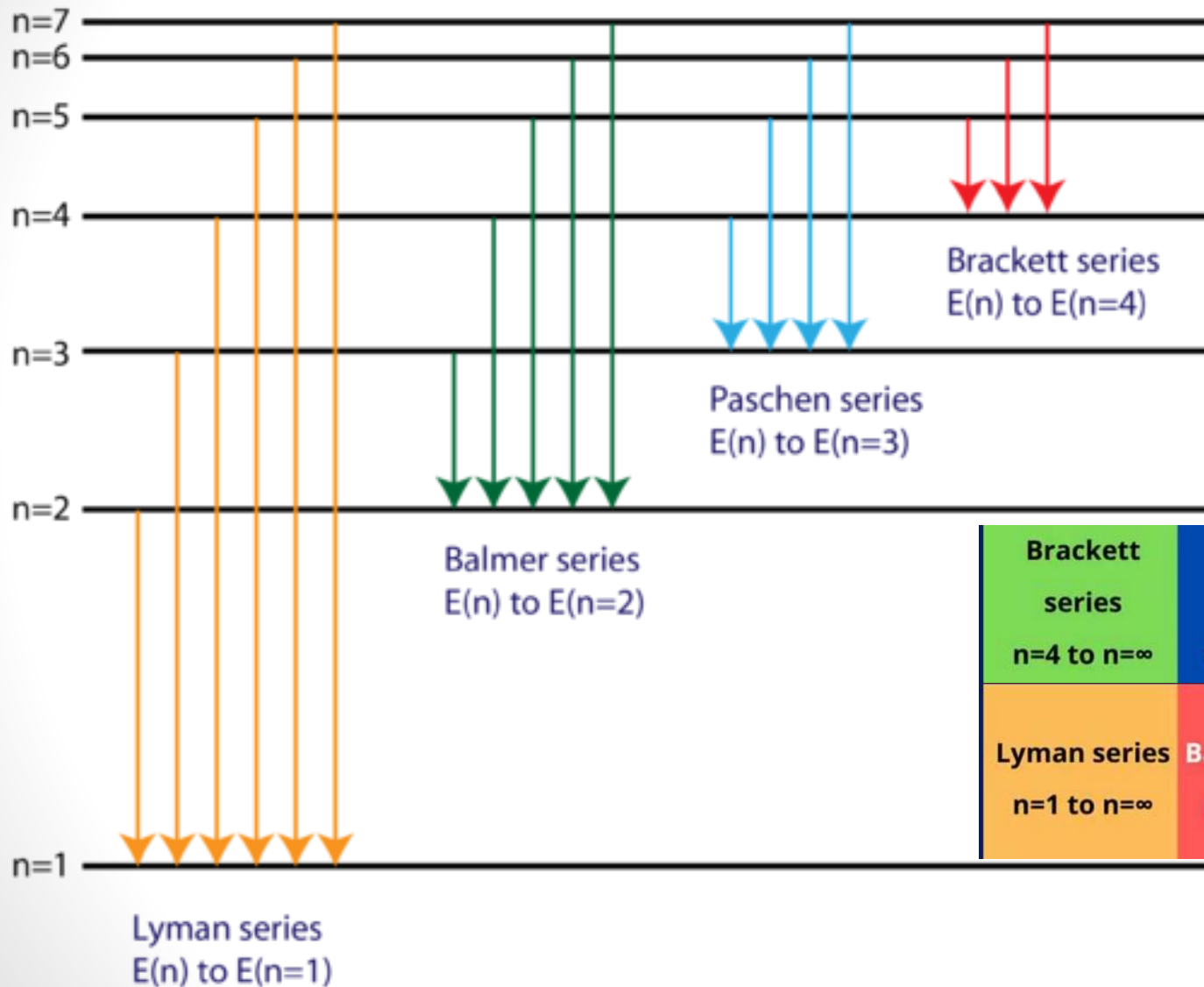
Plank's quantum theory led Einstein , in 1905, in determining the existence of light in form of discrete quanta of energy (photons)

HYDROGEN SPECTRUM



Hydrogen emission spectrum

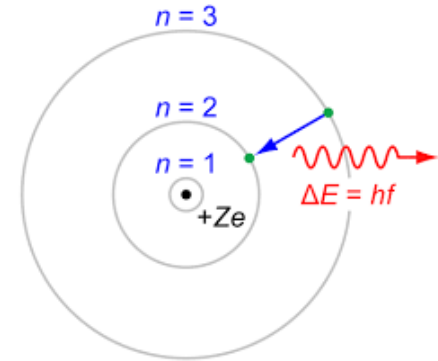
Electron transitions for the Hydrogen atom



Brackett series $n=4$ to $n=\infty$	Pfund series $n=5$ to $n=\infty$	Humphreys series $n=6$ to $n=\infty$
Lyman series $n=1$ to $n=\infty$	Balmer series $n=2$ to $n=\infty$	Paschen series $n=3$ to $n=\infty$

BOHR ATOMIC MODEL:

- Electrons orbits proton (nucleus)in discrete quantized energy states.
- Electrons can transition between energy states either by absorbing or emitting energy (as photons).
- Energy absorbed/ released must be equal to the energy difference between the transition states. ($E = \frac{hc}{\lambda}$)
- Angular momentum of electron is quantized ($L = \frac{nh}{2\pi}$)



Rydberg's Formula:

The Rydberg formula predicts the wavelength of light produced by an electron moving between atomic energy levels. Each element has a unique spectral fingerprint.

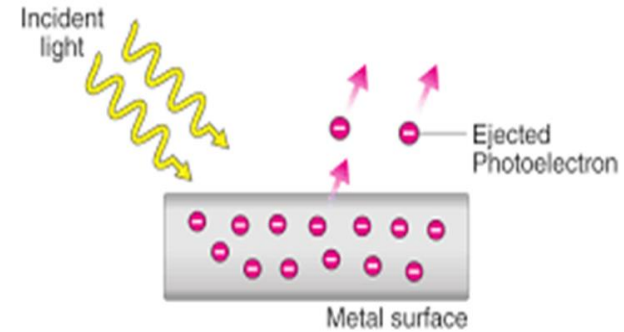
What is R in Rydberg formula?

R is Rydberg constant whose value is equal to $R_{\infty} = [1.0974 \times 10^7 / \text{m}]$

What is formula of Rydberg Formula?

The formula for Rydberg Formula is $1/\lambda = R(1/n_f^2 - 1/n_i^2)$

Photoelectric Effect:

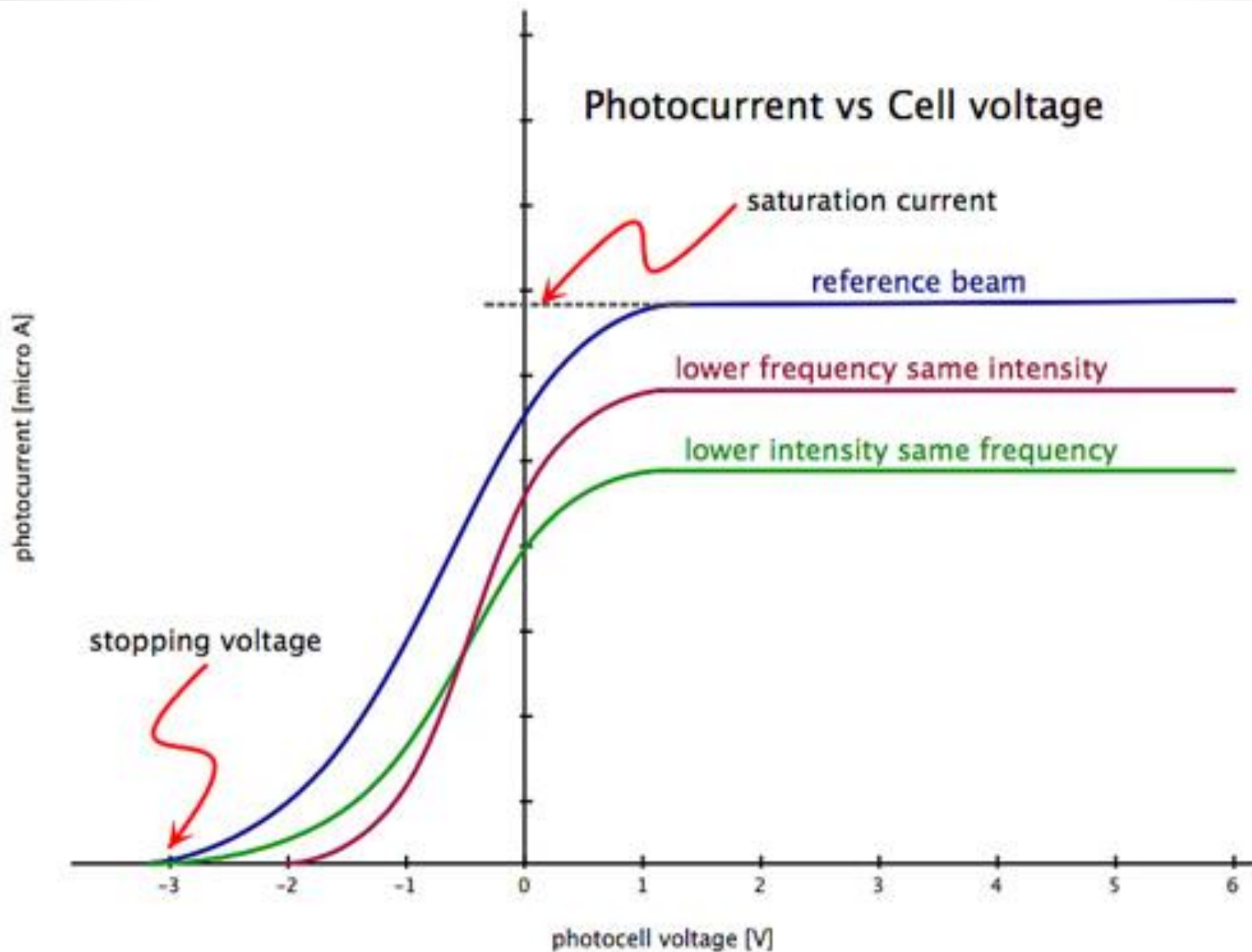


- **Photoelectric effect**, phenomenon in which electrically charged particles are released from or within a material when it absorbs electromagnetic radiation. The **effect** is often defined as the ejection of electrons from a metal plate when light falls on it.

$$\begin{array}{ccccc} \text{Energy of photon =} & & \text{Minimum energy} & & \text{Maximum KE of the} \\ & & \text{needed to remove} & & \text{emitted electron} \\ & & \text{an electron} & & \\ & \nearrow & \nearrow & & \nearrow \\ \boxed{E = hf} & & \boxed{\phi(\text{workfunction})} & & \boxed{E = \frac{1}{2}mv^2} \\ & & & & \\ \boxed{hf = \phi + KE_{\max}} & & & & \end{array}$$

- **THRESHOLD FREQUENCY :**
- The minimum frequency of radiation that will produce a photoelectric effect.
- **PHOTOELECTRIC WORK FUNCTION:**
- The minimum energy required to eject an electron from the surface.
- **STOPPING POTENTIAL:**
- Stopping potential is defined as the potential required to stop ejection of electron from a metal surface when incident beam of energy greater than the work potential of metal is directed on it.

Photocurrent vs Cell voltage



Find the work function (ϕ) of sodium (Na), if the stopping potential of (Na) is $V_0 = 0$ volts and cut off frequency $f_0 = 4.39 \times 10^{14}$ Hz.

Data:

$$V_0 = 0 \text{ volts,}$$

$$f_0 = 4.39 \times 10^{14} \text{ Hz}$$

$$h = \text{plank's constant} = 6.63 \times 10^{-34} \text{ J.s}$$

Solution:

From Einstein Equation of Photoelectric effect,

$$\phi = hf_0 - eV_0 \dots (i)$$

$$\phi = (6.63 \times 10^{-34} \text{ J.s}) (4.39 \times 10^{14} \text{ Hz}) - e(0 \text{ volts})$$

$$\underline{\phi = 2.91 \times 10^{-19} \text{ J}}$$

Converting " ϕ " in electron-volt (eV) from Joule (J),

$$1\text{eV} = 1.602 \times 10^{-19} \text{ Joule}$$

$$\underline{\phi = 1.82 \text{ eV}}$$

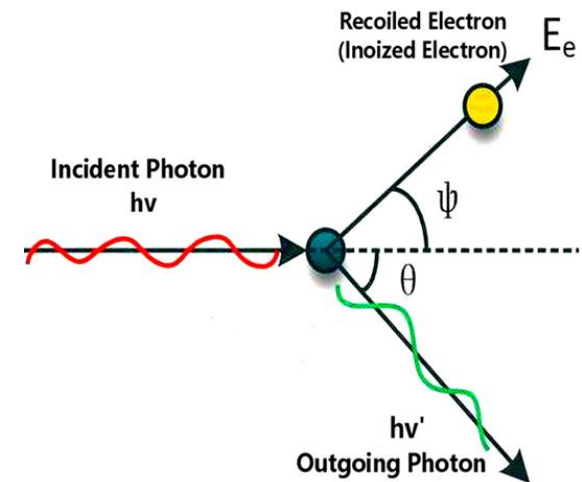
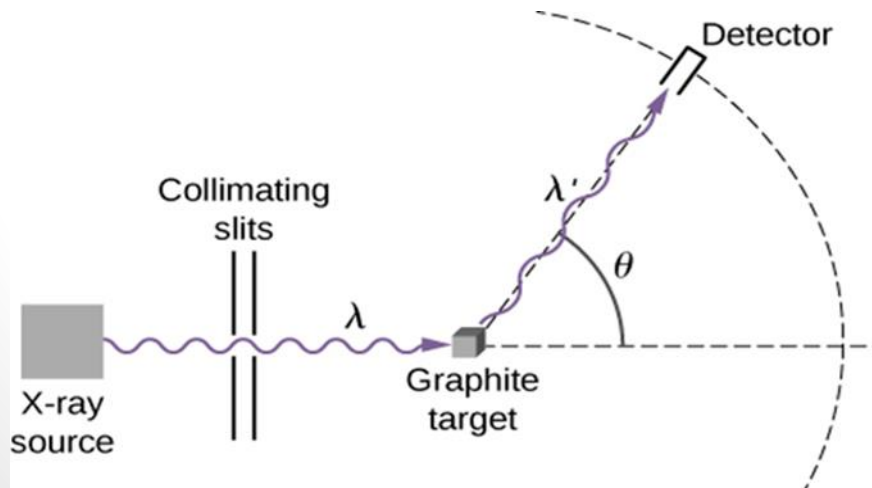
Stopping potential for electron emitted from a surface illuminated by light of wave length 390 nm is 510 mV. When the incident wavelength is changed to a new value, stopping potential is found to be 0.43V. What is the new wave length? And what is the work function?

- Using the formula:
- $\phi = hf_0 - eV_0$

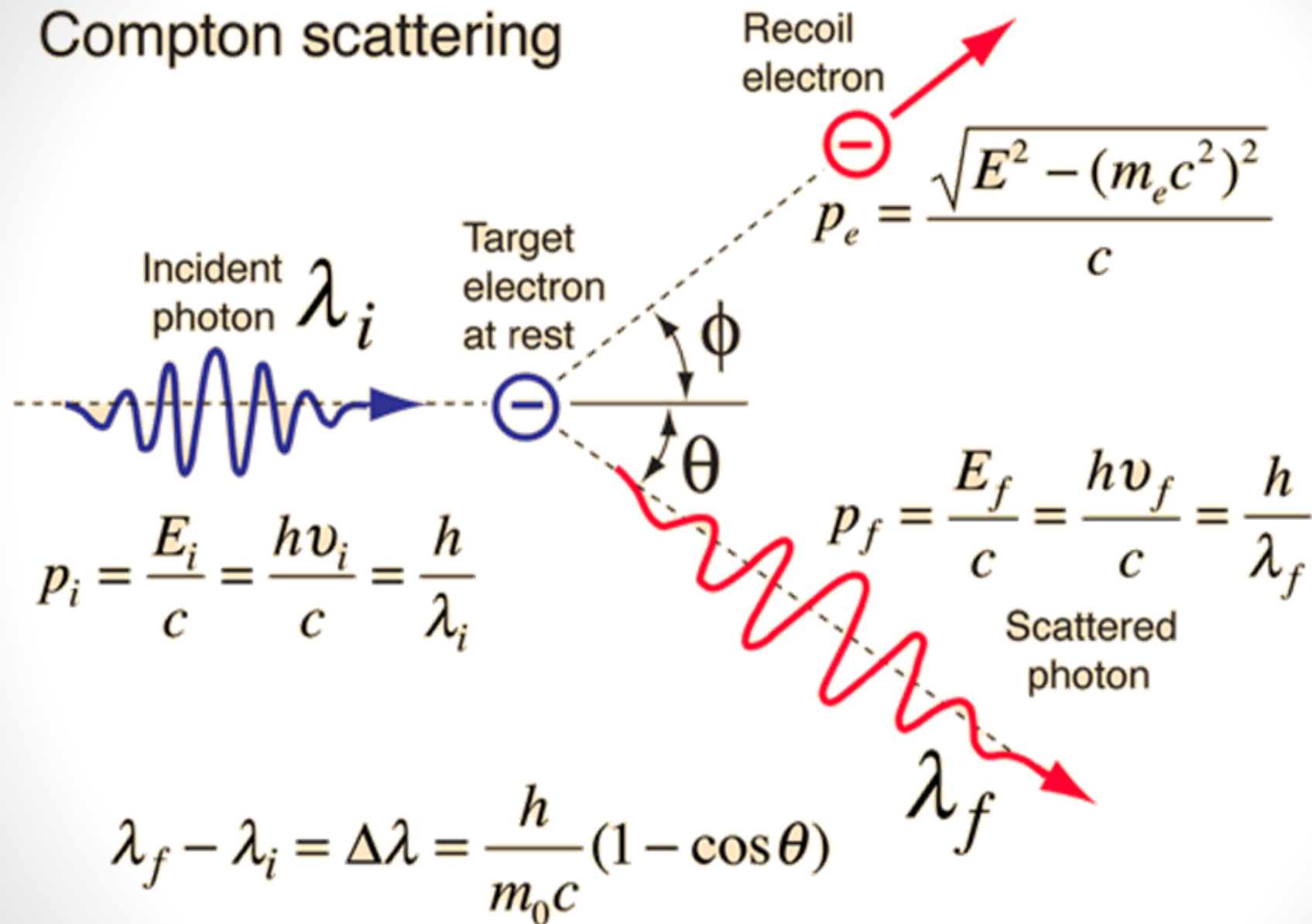
Compton's Scattering:

“The wavelength of Scattered photon is larger than incident photon causing Compton's shift (change in wavelength).”

Compton scattering results when incident monochromatic beam photon (high- energy) collides with a target matter (electron), transferring both the energy (E) and momentum (p) to charge particle of target while the photon moves off with a reduced energy and a change of momentum.



Compton scattering



- Compton effect or Compton scatter is one of principle forms of photon interaction.
- The Compton Effect evidences inelastic collision and in incoherent scattering.
- Compton Effect shows the particle nature of light.
- The amount by which the wavelength of the photon changes is known as **Compton shift**.

A photon of wavelength 2.17 pico-meter are incident on free electrons. CALCULATE,

(a) the wavelength of photon that is scattered at 35° from incident radiation?

(b) wavelength of photon that is scattered at 115° ?

SOLUTION:

From Compton's equation: $\lambda_f - \lambda_i = \Delta\lambda = \frac{h}{m_0 c} (1 - \cos\theta)$

where

λ_i = wavelength of incident photon = 2.17 pico meter

λ_f = wavelength of scattered photon = ?

$$\lambda_i - \lambda_f = \frac{h}{m_e c} (1 - \cos\theta)$$

$$\lambda_i = \lambda_f + \frac{h}{m_e c} (1 - \cos\theta)$$

(a) $\theta = 35^\circ$

$$\lambda_i = (2.17 \times 10^{-12}) + \frac{6.63 \times 10^{-34}}{(9.11 \times 10^{-31})(3 \times 10^8)} (1 - \cos 35^\circ) = 2.6 \times 10^{-12} \text{ m} = 2.6 \text{ pico-meter}.$$

(b) For $\theta = 115^\circ$

$$\lambda_i = (2.17 \times 10^{-12}) + \frac{6.63 \times 10^{-34}}{(9.11 \times 10^{-31})(3 \times 10^8)} (1 - \cos 115^\circ) = 5.62 \times 10^{-12} \text{ m} = 5.62 \text{ pico-meter}$$

De-Broglie Hypothesis:

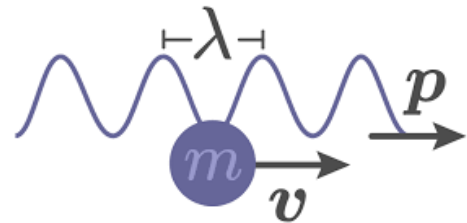
In 1924, French Physicist De-Broglie proposed that:

All matter exhibits wave-like properties and relates the observed wavelength of matter to its momentum.

For example, a beam of electrons can be diffracted just like a beam of light or a water wave

An electron of energy “E” and linear momentum “p” can be described by a matter wave whose wave length and frequency is given by,

$$\lambda = h/p$$



And,

$$f = E/h$$

Where,

h = planks constant

p = linear momentum

E = energy

λ = De Broglie wavelength

ν = frequency

We also know,

$$p = \sqrt{2mK}$$

K = Kinetic Energy

m = mass

So,

$$\lambda = h/\sqrt{2mK}$$

Electron as wave

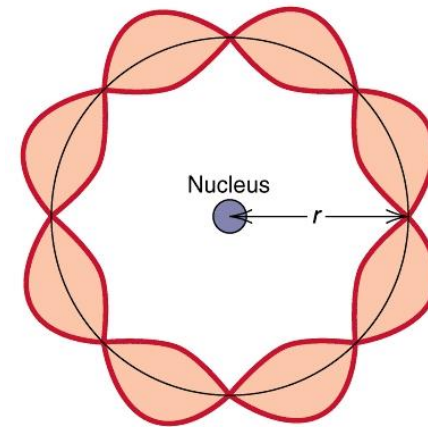
An **electron** of energy “E” and linear momentum “p” can be described by a matter wave

- Only integral numbers of wavelengths are allowed. ($n \lambda$, where $n = 1, 2, 3, \dots$)
- Electrons orbit the nucleus as Stationary/ Standing waves and the circumference ($2\pi r$) of it is quantized.

There for the quantization of orbit:

$$2\pi r = n \lambda$$

Note: Deuterium spectrum put the evidence for De Broglie due to change in momentum of the D hence shift is wavelength (upwards).



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For a hydrogen atom:

Electron wave resonance

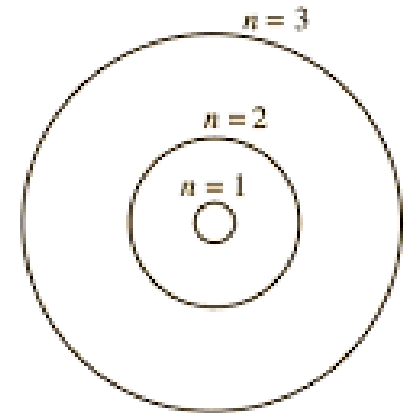
$n = 1$
 $\lambda_1 = 2\pi r_1 = 6.28 a_0$

$n = 2$
 $2\lambda_2 = 2\pi r_2$
 $\lambda_2 = 12.57 a_0$

$n = 3$
 $3\lambda_3 = 2\pi r_3$
 $\lambda_3 = 18.85 a_0$

Wavelengths for hydrogen states.

$a_0 = 0.0529 \text{ nm} = \text{Bohr radius}$



$$r_n = n^2 r_1 = n^2 a_0$$

Uncertainty Principle:

Uncertainty principle, also called **Heisenberg uncertainty principle** or **indeterminacy principle** states that ,

The position and the velocity of an object cannot both be measured exactly, at the same time, even in theory.

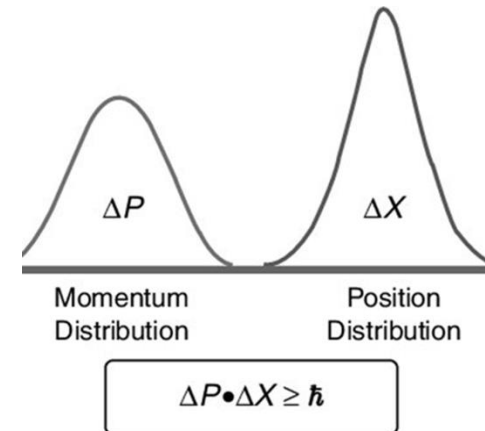
OR

We cannot measure the position (x) and the momentum (p) of a particle with absolute precision. The more accurately we know one of these values, the less accurately we know the other.

The Uncertainty Principle Equation

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$



Uncertainty Relationships

$$\Delta x \cdot \Delta p \sim \hbar$$

$$\Delta E \cdot \Delta t \sim \hbar$$

Precisely determined momentum

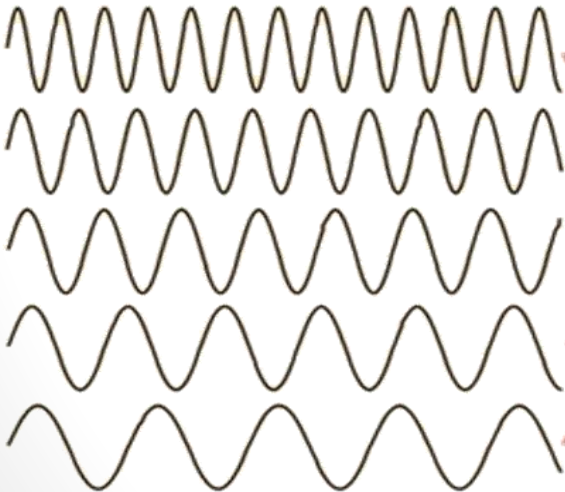


A sine wave of wavelength λ implies that the momentum is precisely known. But the wavefunction and the probability of finding the particle

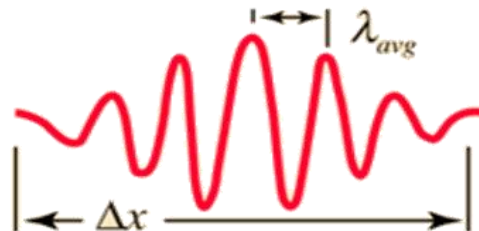
$\Psi^* \Psi$ is spread over all of space!

$$p = \frac{h}{\lambda}$$

p precise
x unknown



Adding several waves of different wavelength together will produce an interference pattern which begins to localize the wave.



But that process spreads the momentum values and makes it more uncertain. This is an inherent and inescapable increase in the uncertainty Δp when Δx is decreased.

$$\Delta x \Delta p > \frac{\hbar}{2}$$

Calculate the De Broglie wavelength of,

- a) dust particle of mass 1.0×10^{-9} kg drifting with a speed of 2.0 cm/s,
- b) an electron whose Kinetic energy is 120 eV.

a) Data:

$$m = 1.0 \times 10^{-9} \text{ kg}$$

$$v = 2.0 \text{ cm/s} = 2.0 \times 10^{-2} \text{ m/s}$$

Solution:

$$\text{De Broglie wavelength : } \lambda = \frac{h}{p} \dots\dots\dots (i)$$

$$\text{For momentum : } p = mv$$

$$\text{Therefore , } \lambda = \frac{h}{mv} = \frac{(6.63 \times 10^{-34} \text{ J.s})}{(1.0 \times 10^{-9} \text{ kg})(2.0 \times 10^{-2} \text{ m/s})}$$

$$\lambda = 3.3 \times 10^{-23} \text{ m}$$

b) Data:

$$K = 120 \text{ eV} = (120 \text{ eV})(1.6 \times 10^{-19} \text{ J/eV}) = 1.92 \times 10^{-17} \text{ J}$$

$$m_e = 9.1 \times 10^{-31} \text{ kg}$$

Solution:

De Broglie wavelength can also be calculated by ,

$$\lambda = \frac{h}{\sqrt{2mK}}$$

$$\lambda = \frac{(6.63 \times 10^{-34} \text{ J.s})}{\sqrt{2(9.1 \times 10^{-31} \text{ kg})(1.92 \times 10^{-17} \text{ J})}}$$

$$\lambda = 1.1 \times 10^{-10} \text{ m}$$

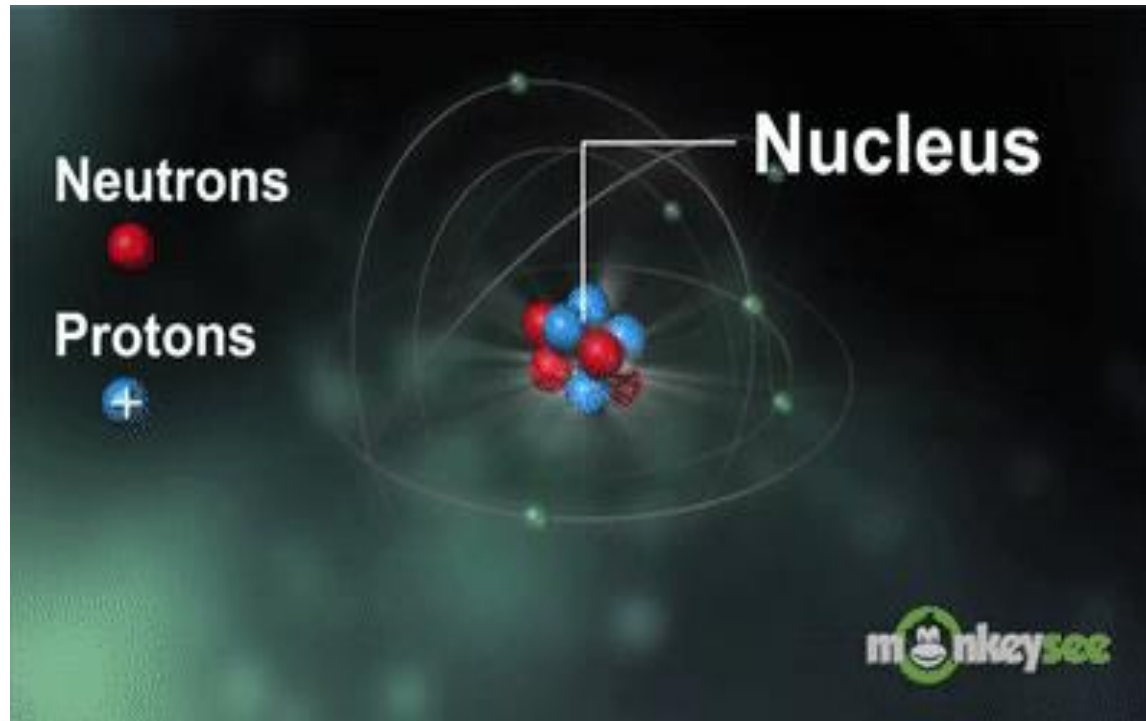
$$\lambda = 1.1 \text{ \AA}$$

Hence, the De Broglie wavelength of electron $\cong$ size of atom.

HOME TASK: A non-relativistic particle is moving three times as fast as an electron. The ratio of the De-Broglie wavelengths, particle to electron is 1.81×10^{-4} . By calculating its mass, identify the particle

Atomic Nucleus:

The **nucleus** is the positively charged center of an **atom** and contains most of its mass. **Nuclei** consist of nucleons (neutrons and protons), the total number of which is equal to the mass number of the **nucleus**.



Nuclear Radiation:

Nuclear radiation is defined as the energy particles or rays that are given off from a radioactive element, such as uranium, as it decays.

- Following are different types of nuclear radiation:
- Alpha radiation – Positively charged (+2) , very large, slow moving and less penetrating radiation. The least dangerous type of radiation. These can be stopped by a sheet of paper.
- Beta radiation – These are fast moving electrons (-). They are highly penetrating; but, it can be stopped by certain metals such as aluminum.
- Gamma radiation - Fast moving and can penetrate everything e.g. X-Rays, except concrete or lead. They are neutral radiation of charge (0).

Characteristics of Some Ionizing Radiations

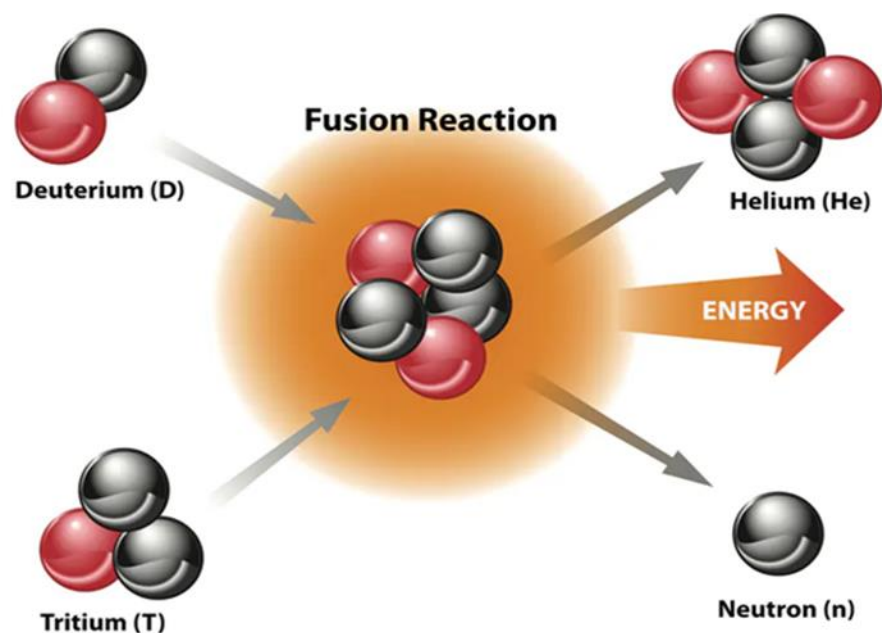
Property	Alpha radiation	Beta radiation	Gamma radiation
Composition	Alpha particle (helium nucleus)	Beta particle (electron)	High-energy electro- magnetic radiation
Symbol	α , He-4	β , e	γ
Charge	2+	1-	0
Mass (amu)	4	$1/1837$	0
Common source	Radium-226	Carbon-14	Cobalt-60
Approximate energy	5 MeV*	0.05 to 1 MeV	1 MeV
Penetrating power	Low (0.05 mm body tissue)	Moderate (4 mm body tissue)	Very high (penetrates body easily)
Shielding	Paper, clothing	Metal foil	Lead, concrete (incomplete shields)

Nuclear Reactions:

Nuclear reactions are processes in which one or more nuclides are produced from the collisions between two atomic nuclei or one atomic nucleus and a subatomic particle.

➤ Nuclear Fusion:

In nuclear fusion reactions, at least two atomic nuclei combine/fuse into a single nucleus. Subatomic particles such as neutrons or protons are also formed as products in these nuclear reactions.

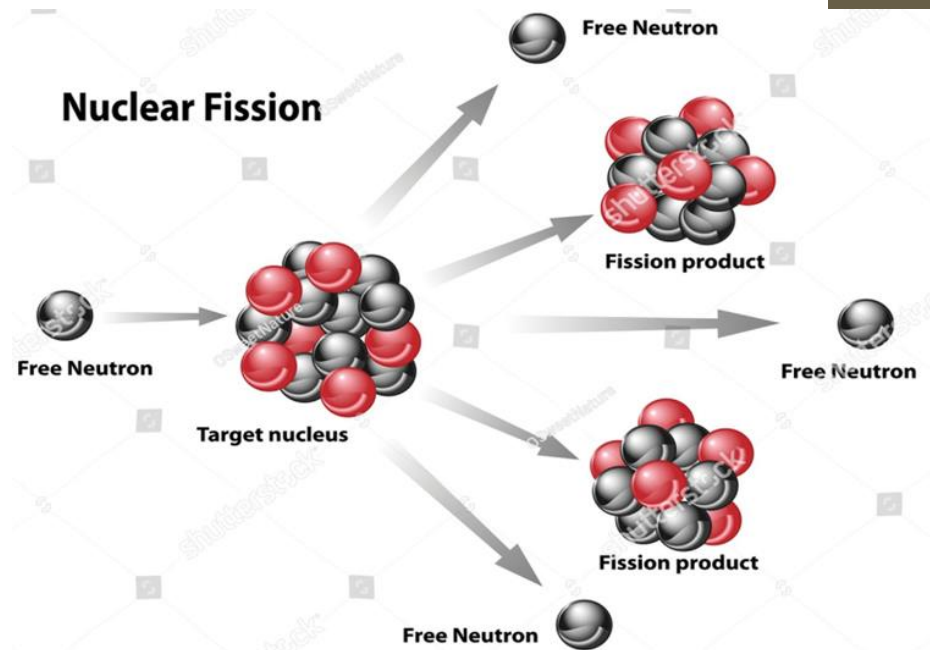
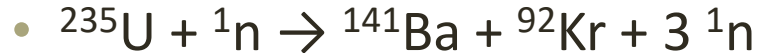


Nuclear Fission:

Nuclear fission refers to the splitting of an atomic nucleus into two or lighter nuclei, through a nuclear reaction or through radioactive decay. It releases a large amount of energy, with the emission of neutrons and gamma rays.

■ *Examples:*

- An important example of nuclear fission is the splitting of the uranium-235 nucleus when it is bombarded with neutrons.



Radioactive Decay and Dating:

“Radioactive Decay is the process in which an unstable nuclei decays (losses its energy) by emitting radiation.”

Carbon-dating:

- The radioactive carbon isotope C^{14} is present in atmosphere with the ratio of $\frac{1}{10^{12}}$.
- Each gram of carbon has radioactivity of 12 decays/ minute. Living organisms can absorb this activity by respiration or by eating plants that have done so. And as organisms die they stop its intake.
- Therefore, by measuring the decay rate of C^{14} , we can determine the age of the sample, this method is known as “radioactive dating”.

$$t = \frac{t_{1/2}}{\ln 2} \ln\left(1 + \frac{Nf}{Ni}\right)$$

- For example: If a sample shows 6 decays per min per gram of carbon, it shows that the original activity have been reduced by half, and the sample must be one half-life (5730 year) old

In a sample of rock, the ratio of Pb^{206} to U^{238} nuclei is found to be 0.65, with a half life of 4.5×10^9 years. what is the average of the rock?

Data:

$$\frac{N_f}{N_i} = 0.65$$

$$t_{1/2} = 4.5 \times 10^9 \text{ years}$$

Solution:

$$t = \frac{t_{1/2}}{\ln 2} \ln\left(1 + \frac{N_f}{N_i}\right) \quad t = \frac{4.5 \times 10^9 \text{ years}}{0.693} \ln(1 +$$